Sodal

An Open Intelligence Platform to

verify human & AI actors, predict behavior, and improve decision making

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Appendix 1: Sodal Overview

Mission, Vision, and Values

Vision

Sodal enhances human agency by enabling fast, safe collaboration for every community

Mission

We believe communities maximize their potential by increasing the agency of each member. Our mission is to deliver technology that enables this freedom. We achieve this by shipping software that creates new, symbiotic markets for agency-advancing capabilities.

At heart, we want to increase human agency. We achieve this by building software the enables four distinct kinds of freedom:

Autonomous Users - Free users to decide who delivers the capabilities they value. We build tools that free users from compulsory dependencies & centralized tethers

Open Products - Free builders to focus on creating differentiated value. We create open systems that drive to zero the cost of capabilities we all need.

Symbiotic Markets - Free producers to capture value through positive-sum synergies rather than zero-sum conflicts. We design products that deliver mutually beneficial outcomes, driving net value creation.

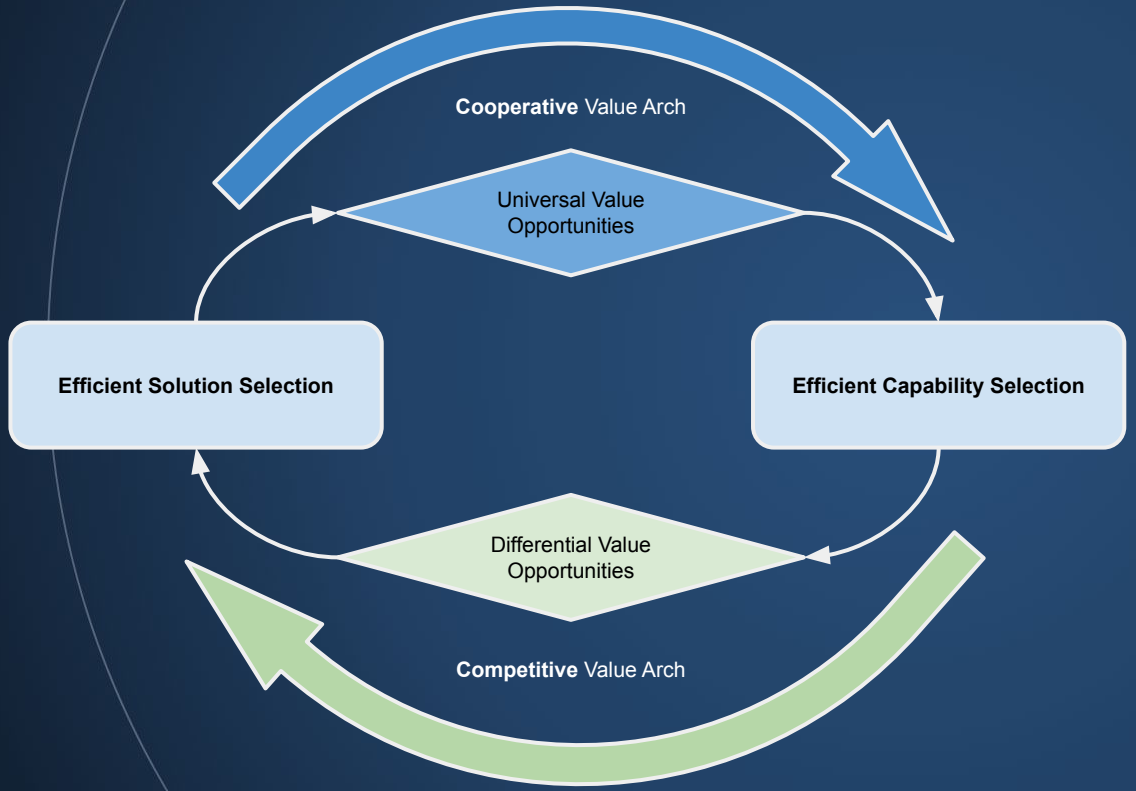
Non-coercive Communities - Free stakeholders to achieve shared goals without enforcing new rules. Our solutions honor the autonomy of others by meeting them where they are.

Strategic Framing

	WHY: Vision	WHAT: Framework	HOW: Technology
GOAL	Enhance Human Agency	Enable Scalable Agentic Tech Adoption	Discover & Coordinate Agency in Digital Systems
KEY CONCEPT	Theory of Value	Model of Agency	Agency Graph Design
ENGINE	Aligned VCs	Profitable Business Platform	Sustainable OSS Community
ACTOR	Ecosystem	Sodal Market	Cartographer Project

Appendix 2: Theory of Value

Theory of Value: Maximize Cooperation + Competition



Truly efficient markets maximize the utility of both ***cooperation*** and ***competition***, recognizing that each draws out a distinct form of excellence that delivers lasting value for the ecosystem.

Universal Value is the name for value generated by the cooperative arch. The pursuit of universal value delivers capabilities that benefit everyone - even competitors.

Differentiated Value is the name for value generated by the competitive arch. The pursuit of differentiated value delivers new solutions, and selects for the best solutions over time.

Undifferentiated Value = Systemic Market Inefficiency

Differentiated Value

Capabilities that stand out as unique or exceptional amid a field of competitors. Market competition for differentiated value improves the ecosystem.

Universal Value

Capabilities that benefit everyone, even competitors, by being broadly accessible. Driving universal value costs to zero improves the ecosystem.

Undifferentiated Value

Capabilities that are not unique or exceptional, but also not broadly available at the lowest cost. Undifferentiated value is an economic and productivity drain on the ecosystem.

The problem with undifferentiated value:

Ecosystems that optimize too aggressively for either *competition* or *cooperation* increase the presence of **undifferentiated value**.

Without competition, cooperative environments become complacent and stagnate. Without cooperation, competitive environments bog down reinventing the wheel, leading to inefficiency and waste.

Both conditions lead to an increase in undifferentiated value: a market of capabilities that are neither particularly exceptional nor particularly accessible.

The SaaS market emphasis on agency-mitigation has measurably increased undifferentiated value

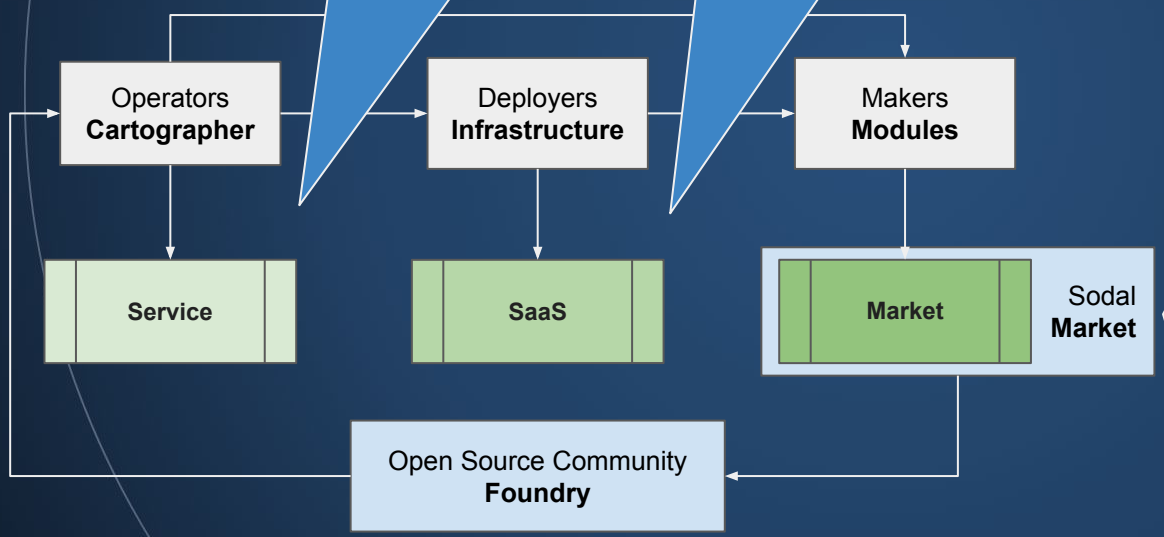
Cooperation + Competition = Open Source + Market *(not SaaS)*

We are not creating a services business

OSS service-based business models have tight margins and difficulty scaling. While we intend Cartographer to be supported by various providers, that is not our business. There is a reason there is only one RedHat.

We are not creating a SaaS business

OSS SaaS becomes a race to trade away your margin, and then a race to trade away your principles. We are not interested in doing either. We intend to maximize OSS' universal value with a distinct differentiated value enabler.



We are creating a Market that scales

Multi-sided markets create highly scalable businesses, delivering differentiated value at the critical juncture between parties.
– *Software* - Apple App Store, Google Play, …
– *Services* - AirBnB, Uber, Lyft, …
– *Content* - TikTok's Influencer Market
– *Gaming* - Valve's Steam Platform

The Sodal Market delivers differentiated value to each participant:

– **Operators want more analytical power.** Give Operators more data integration and analytical power through existing or requested modules

– **Deployers want more service coverage**. Give Deployers a way to choose and implement the best modules to integrate requested data for their existing services, and enable analysis for agentic collaboration goals

– **Module Makers want more revenue opportunities**. Give Module Makers a way to analyze market demand, then create, license, sell, and support modules.

Appendix 3: Model of Agency

Sodal’s Model of Agency

Enhancing agency in our communities involves hard tech, but it is not - at its core - a technology problem.

It is a human problem.

This means that trying to improve agency by emphasizing a specific technical intervention - *a tactical “how”* - is unlikely to be successful on its own. Rather, we need a framework for understanding the nature of agency in digital systems. Only then can we assess what we must do (or not) to improve agency.

Sodal’s approach starts with a comprehensive model of human agency, and then uses that model to map how various technical systems can be adapted to enhance individual agency.

We represent this model as a 3 x 3 grid, *a 9-box*, that progresses from “initial awareness of the systems we depend on” to a comprehensive understanding of actions we can take to generate new opportunities for desired collaboration.

The 9-box forms a **map of market opportunities to be engaged as modules**.

Model Axes: Actions & Objects

Action Ladder (horizontal axis)

The action ladder describes the three stages of control we exercise in our environment. It is a ladder, with the higher building on the lower - we cannot administer systems we haven't discovered, or control our presence without the ability to first administer capabilities.

3 – Presence:
controlling how other participants identify and interact with me

2 – Administration:
managing capabilities I own or control

1 – Discovery:
identifying systems, data, and participants that I interact with

Object Ladder (vertical axis)

The object ladder describes the three stages of agent realization in digital environments. All digital presence is realized by an instantiating system, and is defined by various data. Like actions, each higher "rung" depends on those beneath, culminating in a definable *participant*.

3 – Participants:
actors known to exercise agency through system capabilities

2 – Data:
record of capability configurations, content, and identifiers

1 – Systems:
source of being for capabilities, content, and identifiers

The 9-box Model of Agency

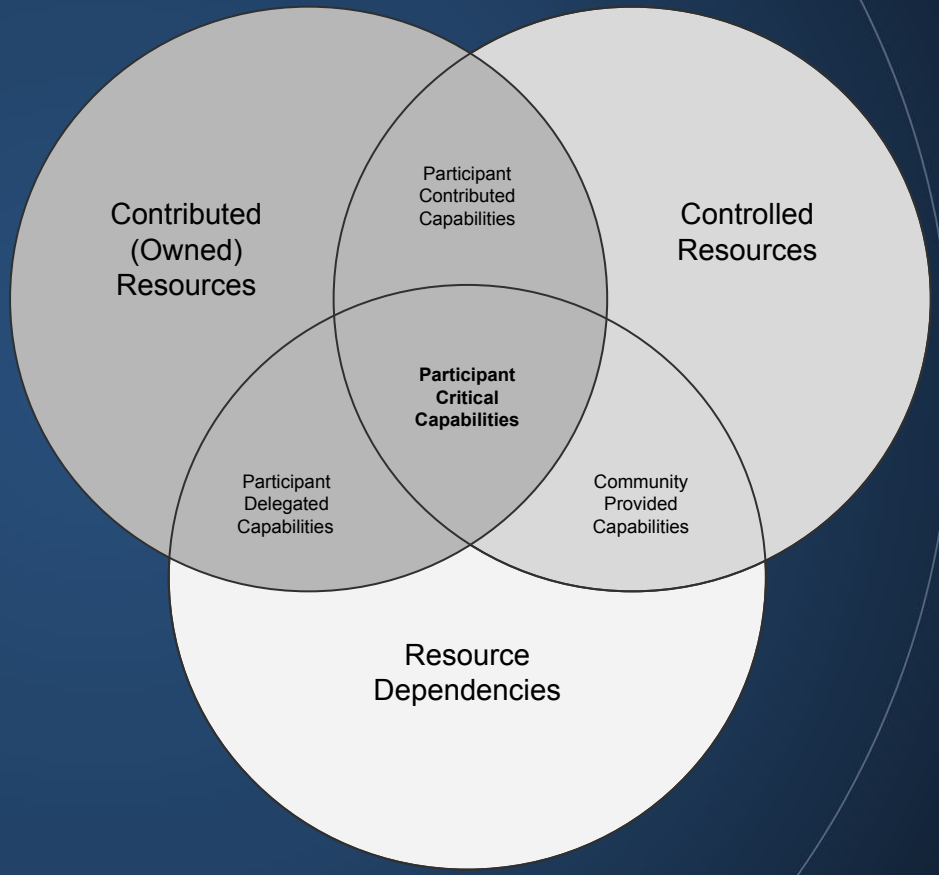
	Discovery *How I discover and interact with other people*	Administration *How I manage capabilities I own or control*	Presence *How participants discover and interact with me*
Participants *nondeterministic agents*	**Participant Discovery** I model agency of all known participants *in order to increase* **participant behavioral certainty**	**Participant Administration** I configure participant access *in order to increase* **desired collaboration**	**Participant Presence** I manage associated access and behaviors *in order to increase* **opportunities for desired collaboration**
Data *personas & configuration*	**Data Discovery** I model data in my environment *in order to increase* **agency insights**	**Data Administration** I manage data I control *in order to increase* **probability of desired collaboration**	**Data Presence** I manage data linked to my identifiers *in order to increase* **control over insights into my agency**
Systems *instantiating platforms*	**Systems Discovery** I model system capabilities in my environment *in order to increase* **data to produce insights**	**Systems Administration** I configure systems I control *in order to increase* **system behavioral certainty**	**Systems Presence** I manage system identifiers associated with me *in order to increase* **control of data linked to my identifiers**

Appendix 4: Agency Graph Design

Preamble: Defining Shared Resources

Agency graphs address the challenge of operating in a shared resource environment. Shared resource environments have properties that do not lend themselves to centralized service management, i.e.:

- Participants may not own the resources they use
- Participants frequently control resources they do not own
- Participants share resources with others
- Participants depend on resources they neither own or control



Agency Graph Introduction

The core of our solution is a probabilistic graphical model of agency in digital systems called a **agency graph**. Person graphs enable **shared resource** management by clarifying and managing how influence flows through interconnected digital systems.

Agency graphs use **composite identities** called **personas** to model how participants influence the use of resources within digital systems. We refer to this influence as **agency**.

Agency graphs model agency by describing the relationships between personas, illuminating when one persona can act on behalf of another persona. The result is a queryable graph of agency informed by every connected digital system.

Agency Graphs are probabilistic graphical models of agency in digital systems

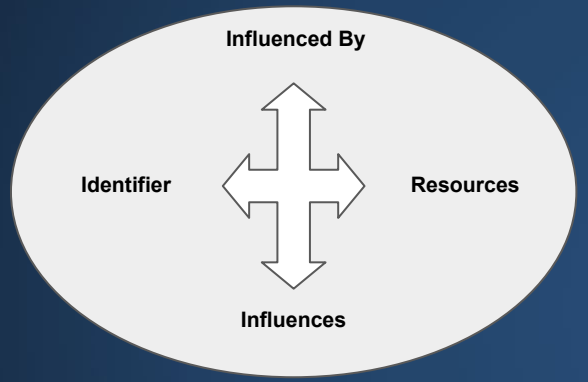
Composite Identities are identities that unite identifiers, capabilities, actions, and relationships

Behavioral Certainty is a measure of our confidence in an agent's expected behavior

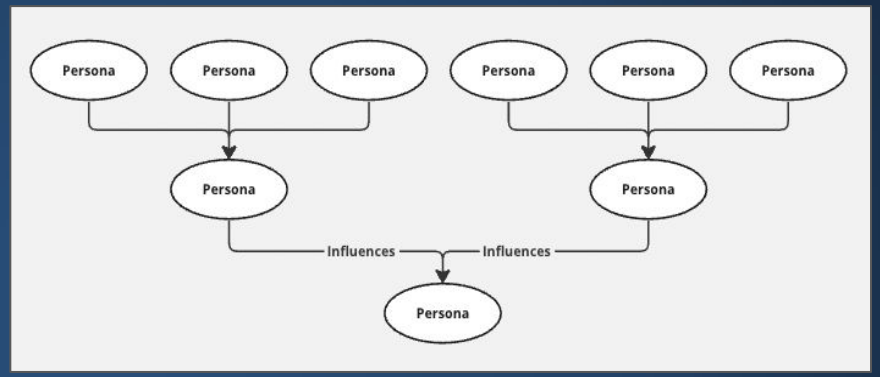
Dynamic Agency describes how new data changes our understanding of influence

Implicit Roles are roles that emerge organically from our understanding of a graph

Persona nodes represent composite identities



Persona



Agency Graph

A **persona** is an inclusive concept. As we extend and improve the graph, it modifies our understanding of all connected nodes, a quality we call **dynamic agency**. The more robust our graph becomes, the more confident we can be about the **expected behavior** of personas, increasing **behavioral certainty**. Finally, we can use this understanding to identify places we can influence to achieve the greatest probability of an increase in **desired behavior** of the entire system.

Graph edges express the flow of agency

Agency graph edges contain three elements: a **direction**, a **control level**, and a **confidence score**.

confidence score ranges from 0 to 1, indicating the operator's confidence in the edge's assertion.

control level describes the type of agency expressed by the edge (*the table lists the 12 valid control levels*)

	Will *compel to act*	**Be** *cause to exist*	**Do** *cause to act*
root	**possess** (have physical control)	**alias** (is presentation of)	**admin** (unrestricted control)
execute	**direct** (have immediate authority)	**realize** (cause to exist right now)	**manage** (change which nodes can interact)
write	**govern** (have indirect authority)	**define** (establish nature of being)	**act_as** (cause to interact with other nodes)
read	**inform** (introduce authority considerations)	**describe** (communicate nature of being)	**access** (cause to interact with my node)